

# Chapter 4

## STAT 5700: Probability

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Spring 2026

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## 4.2 Continuous Random Variables

### Discrete vs. Continuous

So far we have worked with **discrete** random variables that have probability distributions

$$\sum_S p(y) = 1$$

In this chapter we will consider **continuous** random variables that have **probability density functions** (pdfs)

$$\int_S f(y)dy = 1$$

Discrete random variables have a finite or countably infinite set of possible outcomes:

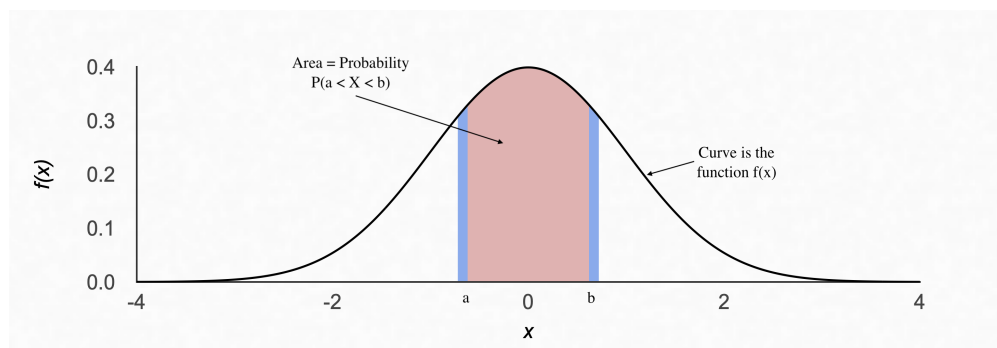
- Uniform (m-sided die):  $\{1, 2, \dots, m\}$
- Bernoulli (success/failure):  $\{0, 1\}$
- Binomial (# of successes in n trials):  $\{0, 1, \dots, n\}$
- Geometric (# of trials until 1st success):  $\{1, 2, \dots\}$
- Negative Binomial (# of trials until rth success):  $\{r, r+1, r+2, \dots\}$
- Hypergeometric (# of successes when sampling w/o replacement from finite pop):  $\{0, 1, 2, \dots, r\}$
- Poisson (# of occurrences in unit interval)  $\{1, 2, \dots\}$

**Continuous** random variables, on the other hand, have an *interval* of possible outcomes, and decimal values are possible:

- amount of rain, in inches, that falls in a randomly selected storm
- weight, in pounds, of a randomly selected student
- square footage of a randomly selected house

The set of possible measurements in a continuous interval is (not countably) infinite and can't be put on a one-to-one correspondence with the integers.

The probability that a continuous random variable  $Y$  takes on any particular value is 0, so we won't be finding  $P(Y = y)$  for continuous random variables, but rather,  $P(a < Y < b)$ .

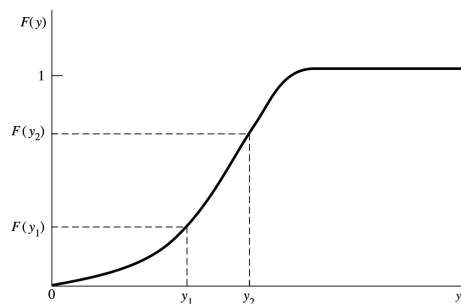


## Recall: cumulative distribution functions (cdfs)

$$F(y) = P(Y \leq y), \quad -\infty < y < \infty$$

is called the **cumulative distribution function (cdf)** of a random variable  $Y$ .

FIGURE 4.2  
Distribution function  
for a continuous  
random variable



**Theorem 4.1:** Properties of a cumulative distribution function: If  $F(y)$  is a cdf, then

1.  $F(-\infty) \equiv \lim_{y \rightarrow -\infty} F(y) = 0$ .
2.  $F(\infty) \equiv \lim_{y \rightarrow \infty} F(y) = 1$ .
3.  $F(y)$  is a nondecreasing function of  $y$ . If  $y_1 < y_2$ , then  $F(y_1) \leq F(y_2)$ .

## Properties of continuous pdfs and cdfs

| Discrete                                              | Continuous                                                                        |
|-------------------------------------------------------|-----------------------------------------------------------------------------------|
| $p(y)$ = probability distribution                     | $f(y)$ = probability density function (pdf)                                       |
| $p(y) > 0, y \in S$                                   | $f(y) \geq 0, y \in S$                                                            |
| $\sum_{y \in S} p(y) = 1$                             | $\int_S f(y) dy = 1$                                                              |
| If $A \subset S$ , $P(Y \in A) = \sum_{Y \in A} p(y)$ | If $(a, b) \subset S$ ,<br>$P(a < Y < b) = P(a \leq Y \leq b) = \int_a^b f(y) dy$ |
| cdf: $F(y) = P(Y \leq y)$                             | cdf: $F(y) = P(Y \leq y) = \int_{-\infty}^y f(t) dt, -\infty < y < \infty$        |

By the **Fundamental Theorem of Calculus**, we have that  $F'(y) = f(y)$  for all  $y$  values where  $F'(y)$  exists, and  $P(a \leq Y \leq b) = F(b) - F(a)$

**Example:** Let  $Y$  be a continuous random variable with pdf given by

$$f(y) = \begin{cases} 3y^2, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Verify that  $f(y)$  is a valid pdf and find  $F(y)$ .

**Example:** Given  $f(y) = cy^2, 0 \leq y \leq 2$ , find the value of  $c$  for which  $f(y)$  is a valid pdf. Then, find  $P(1 \leq Y \leq 2) = P(1 < Y < 2)$

### 4.3 Expected value of continuous distributions

Expected value (including mean, variance, and moment generating functions) have the *same exact* definition for continuous random variables as they did for discrete random variables. Now, we're just evaluating them with **integrals** instead of **sums**.

| Definition                         | Discrete                          | Continuous                  |
|------------------------------------|-----------------------------------|-----------------------------|
| $\mu = E(Y)$                       | $\sum_{y \in S} yp(y)$            | $\int_S yf(y)dy$            |
| $\sigma^2 = V(Y) = E[(Y - \mu)^2]$ | $\sum_{y \in S} (y - \mu)^2 p(y)$ | $\int_S (y - \mu)^2 f(y)dy$ |
| $k^{th} \text{ moment} = E(Y^k)$   | $\sum_{y \in S} y^k p(y)$         | $\int_S y^k f(y)dy$         |
| $m(t) = E(e^{tY})$                 | $\sum_{y \in S} e^{ty} p(y)$      | $\int_S e^{ty} f(y)dy$      |
| $E(g(Y))$                          | $\sum_{y \in S} g(y)p(y)$         | $\int_S g(y)f(y)dy$         |

The following relationships still hold true for continuous random variables as well:

- standard deviation =  $\sigma = \sqrt{\sigma^2} = \sqrt{V(Y)}$
- $V(Y) = E(Y^2) - [E(Y)]^2$
- first moment:  $\mu = E(Y) = m'(0)$
- second moment:  $E(Y^2) = m''(0)$
- $\sigma^2 = V(Y) = m''(0) - [m'(0)]^2$
- $E(cg(Y)) = cE(g(Y))$
- $E[c_1g_1(Y) + c_2g_2(Y)] = c_1E(g_1(Y)) + c_2E(g_2(Y))$

## 4.4 Uniform Distribution (continuous)

### Example

Suppose it takes between 0 and 40 seconds for an elevator to arrive once you have pushed the button. We will assume that all wait times are equally likely. Define a random variable and determine a pdf to model this situation. *Hint: draw a picture.*

**Example**

Let the random variable  $Y$  denote the outcome when a point is selected at random from an interval  $[\theta_1, \theta_2]$ ,  $-\infty < \theta_1 < \theta_2 < \infty$ . What might be a reasonable expression for the probability that the point is selected from the interval  $[\theta_1, y]$ ,  $\theta_1 \leq y \leq \theta_2$ . *Hint: draw a picture.*

The *cdf* of a **continuous uniform random variable**  $Y$  is given by

$$F(y) = \begin{cases} 0, & y < \theta_1 \\ \frac{y - \theta_1}{\theta_2 - \theta_1}, & \theta_1 \leq y \leq \theta_2, \\ 1, & \theta_2 \leq y, \end{cases}$$

and the *pdf* of  $Y$  is given by

$$f(y) = \frac{1}{\theta_2 - \theta_1}, \quad \theta_1 \leq y \leq \theta_2.$$

Similar to the discrete case, a random variable  $Y$  has a **uniform distribution** if its pdf is equal to a constant on its support.

However, in the continuous case, the support is a continuous interval rather than a discrete set of possible values. In the definition on the previous slide, the support of  $Y$  is the interval  $[\theta_1, \theta_2]$ .

We write  $Y \sim U(\theta_1, \theta_2)$ , to mean  $Y$  is a uniform random variable (or “is uniformly distributed”) on the interval  $(\theta_1, \theta_2)$ .

You will derive the mean, variance, and mgf of the continuous uniform distribution in the group work.

Note that we often consider the uniform distribution in conjunction with a Poisson process. For example, if the number of events, such as calls coming into a switchboard, that occur in an interval  $(0, t)$  has a Poisson distribution, and it is known that exactly one such event occurred in that interval, then the actual time of occurrence is distributed uniformly over this interval.

**Elevator Example (cont'd)**

Find and graph the cdf for the uniform random variable in the elevator example. What is the probability that the elevator arrives in less than 10 seconds? Between 15 and 20 seconds?

**Exercise**

Let  $Y$  have a uniform distribution  $U(0, 1)$  (we call this the “standard uniform distribution”).

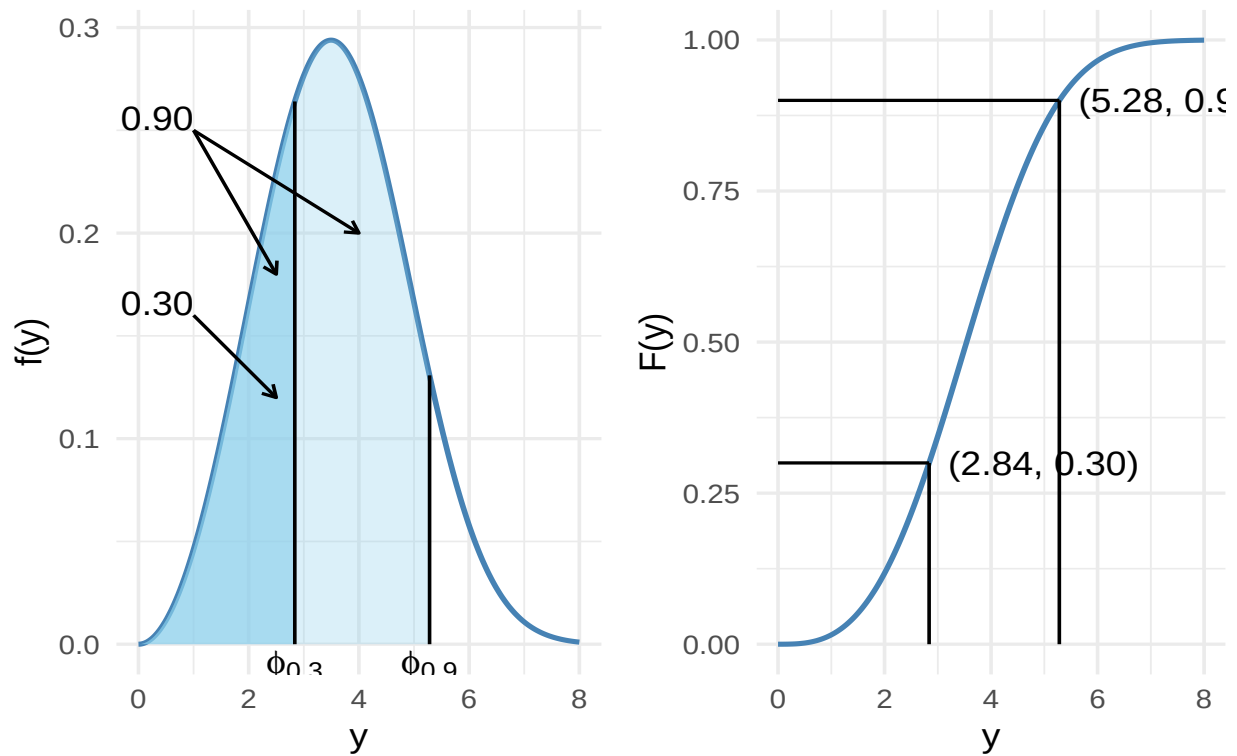
1. Write out the pdf and cdf of  $Y$
2. Let  $W = a + (b - a)Y$ ,  $a < b$  Find the cdf of  $W$ . How is  $W$  distributed?

## Percentiles

The **(100p)th percentile** is a number  $\phi_p$ , such that the area under  $f(y)$  to the left of  $\phi_p$  is  $p$ . That is,

$$F(\phi_p) = \int_{-\infty}^{\phi_p} f(y) dy = p$$

For example, the figure below illustrates the 30th ( $\phi_{0.3}$ ) and 90th ( $\phi_{0.9}$ ) percentiles.



The 50th percentile is called the **median**,  $\phi_{0.50}$ . The 25th and 75th percentiles are called the **first** and **third quartiles**, respectively, and are denoted by  $q_1 = \phi_{0.25}$  and  $q_3 = \phi_{0.75}$ .



**Exercise**

The time  $Y$  in months until the failure of a certain product has pdf and cdf

$$f(y) = \frac{3y^2}{4^3} e^{-(y/4)^3}, \quad 0 < y < \infty$$

$$F(y) = \begin{cases} 0, & -\infty < y < 0, \\ 1 - e^{-(y/4)^3} & 0 \leq y < \infty \end{cases}$$

This is the distribution depicted in the figure above.

**Find the median of this distribution.**

## 4.5 The Normal Distribution

The normal distribution has two parameters,  $\mu$  and  $\sigma^2$ . We write  $Y \sim N(\mu, \sigma^2)$ .  $Y$  has a normal distribution if its pdf is defined by:

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y-\mu)^2}{2\sigma^2}}, \quad -\infty \leq y < \infty,$$

and the mgf is given by:

$$m(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

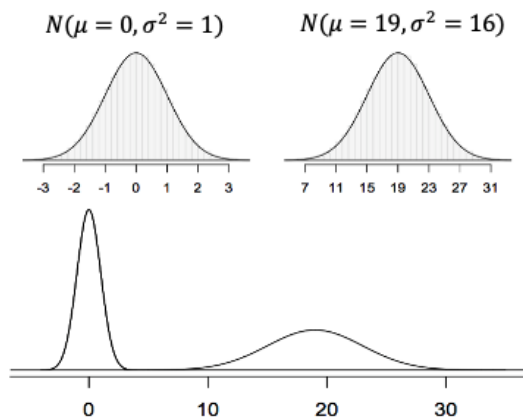
### Normal mean and variance

Use the mgf to find the mean and the variance of the Normal distribution.

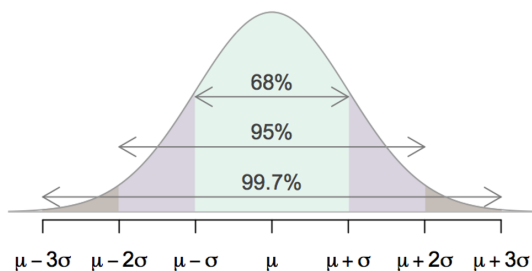
The normal distribution is unique in that its parameters in the pdf ( $\mu$  and  $\sigma^2$ ) are the same as its mean and variance.

This is NOT necessarily true for other distributions. For example, the Uniform distribution is defined by the two parameters  $\theta_1$  and  $\theta_2$ , that is  $Y \sim U(\theta_1, \theta_2)$ , but  $E(Y) = \frac{1}{2}(\theta_1 + \theta_2) \neq \theta_1$  and  $V(Y) = \frac{1}{12}(\theta_2 - \theta_1)^2 \neq \theta_2$

## Properties of the Normal Distribution



- The normal distribution is unimodal and symmetric.
- The parameter  $\mu$  controls the *center* of the distribution, and the parameter  $\sigma^2$  controls the *spread* of the distribution.
- The normal distribution has a convenient property known as the **Empirical Rule**:
  - roughly 68% of the probability falls within  $1\sigma$  of the mean
  - roughly 95% falls within  $2\sigma$  of the mean
  - roughly 99.7% falls within  $3\sigma$  of the mean



### Exercise:

SAT scores are distributed nearly normally with mean 1000 and standard deviation 200.

~68% of students score between \_\_\_\_\_ and \_\_\_\_\_ on the SAT.

~95% of students score between \_\_\_\_\_ and \_\_\_\_\_ on the SAT.

~99.7% of students score between \_\_\_\_\_ and \_\_\_\_\_ on the SAT

## Standard Normal Distribution (Z)

The normal distribution with  $\mu = 0$  and  $\sigma^2 = 1$  is known as the **standard normal distribution**, and we usually use  $Z$  instead of  $Y$  to denote a standard normal random variable:  $Z \sim N(0, 1)$ .

The cdf of the standard normal distribution is also given special notation:

$$\Phi(z) = F(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-w^2/2} dw$$

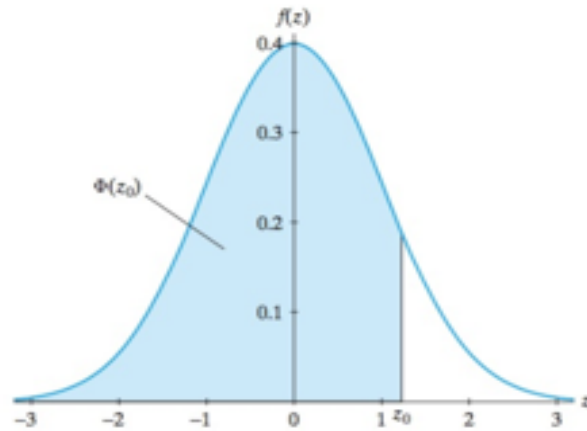


Figure 3.3-1 Standard normal pdf

We can convert any normal distribution into a standard normal distribution.

Let  $Y \sim N(\mu, \sigma^2)$  and define the random variable  $Z = \frac{Y - \mu}{\sigma}$

1. Show that  $E(Z) = 0$

2. Show that  $V(Z) = 1$

That is, the mean and the variance of  $Z$  are 0 and 1, respectively, no matter what the mean and variance of  $Y$ .

**Theorem** If  $Y \sim N(\mu, \sigma^2)$ , then  $Z = (Y - \mu)/\sigma$  is  $N(0, 1)$

**Z-scores** are a way of understanding data as distances from the mean in “standard deviation units”

### Properties of Z-scores

- Z-scores follow a standard normal distribution  $Z \sim N(0, 1)$
- Z-scores will always be centered at \_\_\_\_\_
- Observations that fall BELOW the mean will have a \_\_\_\_\_ z-score
- Observations that fall ABOVE the mean will have a \_\_\_\_\_ z-score
- Observations that are more than 2 SD away from the mean ( $|Z| > 2$ ) are usually considered unusual.
- Nearly all Z scores will be between \_\_\_\_\_ and \_\_\_\_\_ due to the \_\_\_\_\_

## 4.6 The Exponential, Gamma, and Chi-square Distributions

A Poisson distribution is a discrete distribution used to model the number of occurrences in a given time interval. In particular, let  $Y$  be the number of occurrences in an interval of length  $t$ , then  $Y$  is a Poisson random variable with pmf

$$f(y) = \frac{(\lambda t)^y e^{-\lambda t}}{y!}, \quad y = 0, 1, 2, \dots$$

$\lambda$  is interpreted as the mean number of occurrences in the unit interval  $[0, 1]$ . The number of occurrences is a discrete random variable, but the waiting times between successive occurrences are also random variables. However, these waiting times are continuous.

#### Exercise

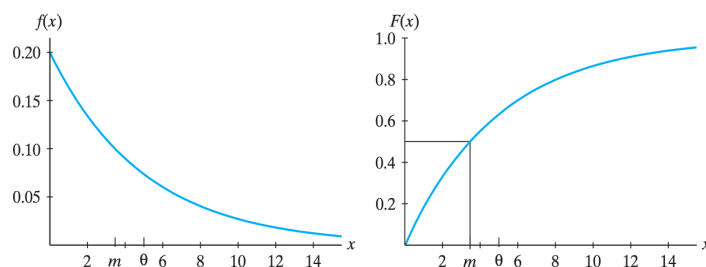
Let  $W$  denote the waiting time until the first occurrence for a Poisson process in which the mean number of occurrences in the unit interval is  $\lambda$ .

What is the probability the time until first occurrence is less than  $w$ ?

Use the above result to produce the pdf of  $W$

## Exponential Distribution

We just derived the pdf for what's known as the **exponential distribution**, which is used to model the waiting time for a first occurrence. We usually re-parameterize and let  $\lambda = \frac{1}{\beta}$ , because doing so gives the convenient interpretation of  $\beta$  as the mean waiting time,  $E(Y)$ . We will derive this result soon.



**Figure 3.2-1** Exponential pdf,  $f(x)$ , and cdf,  $F(x)$

$Y$  has an **exponential distribution** if its pdf is defined by

$$f(y) = \frac{1}{\beta} e^{-y/\beta}, \quad 0 \leq y < \infty$$

and the cdf is defined by

$$F(y) = \begin{cases} 0 & y < 0 \\ 1 - e^{-y/\beta} & 0 \leq y < \infty \end{cases}$$

We say  $Y \sim \exp(\beta)$ .

Its mgf, mean, and variance are given by:

$$m(t) = \frac{1}{1 - \beta t}, \quad t < \frac{1}{\beta}$$

$$E(Y) = \beta$$

$$V(Y) = \beta^2$$

Use the mgf to derive the mean and variance of the exponential distribution. That is, show that for  $Y \sim \exp(\beta)$ , then  $E(Y) = \beta$  and  $V(Y) = \beta^2$ .

**Example**

Customers arrive in a certain shop according to an approximate Poisson process at a mean rate of 20 per hour (i.e.  $1/3$  per minute). What is the probability that the shopkeeper will have to wait more than 5 minutes for the arrival of the first customer? Also find the median time until the first arrival.



## Gamma Distribution

The **Gamma distribution** is used to model the time until the  $\alpha$ th occurrence in a Poisson process (with mean  $\lambda$  arrivals per unit.) The distribution has two parameters,  $\alpha$  (the number of occurrences) and  $\beta = 1/\lambda$ . We write  $Y \sim \text{gamma}(\alpha, \beta)$  or sometimes  $Y \sim \Gamma(\alpha, \beta)$  using the capital Greek letter Gamma ( $\Gamma$ ).

$$f(y) = \frac{1}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1} e^{-y/\beta}, \quad 0 \leq y < \infty$$

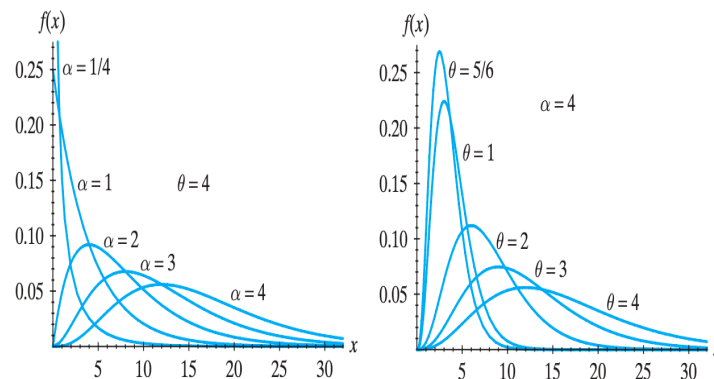
$$m(t) = \frac{1}{(1 - \beta t)^\alpha}, \quad t < 1/\beta$$

$$E(Y) = \alpha\beta$$

$$V(Y) = \alpha\beta^2$$

The pdf of a gamma random variable involves what's known as the **gamma function**:  $\Gamma(t) = \int_0^\infty y^{t-1} e^{-y} dy$ ,  $0 < t$

When  $n$  is a positive integer, it can be shown that  $\Gamma(n) = (n-1)!$ . Typically, we will use R to evaluate the gamma function, particularly when  $n$  is not an integer.



**Figure 3.2-2** Gamma pdfs:  $\theta = 4, \alpha = 1/4, 1, 2, 3, 4$ ;  $\alpha = 4, \theta = 5/6, 1, 2, 3, 4$

**Example**

Suppose the number of customers per hour arriving at a shop follows a Poisson process with mean 30 people per hour. What is the distribution required to determine the probability the shopkeeper will wait more than 5 minutes before both of the first two customers arrive? What is the mean and variance of the random variable?

**The Chi-square distribution**

A special case of the **Gamma** distribution with  $\alpha = \nu/2$  and  $\beta = 2$  is known as the **Chi-square distribution**, where  $\nu$  is a positive integer.

Let  $Y \sim \Gamma(\nu/2, 2)$ . The pdf of  $Y$  is

$$f(y) = \frac{1}{\Gamma(\nu/2)2^{\nu/2}} y^{\nu/2-1} e^{-y/2}, \quad 0 \leq y < \infty.$$

We say that  $Y$  has a **chi-square distribution with  $\nu$  degrees of freedom**, which we abbreviate as  $Y \sim \chi^2(\nu)$

**Exercise**

Using the properties of the gamma distribution, find the mean and variance of the chi-square distribution.

**Theorem** If the random variable  $Y$  is  $N(\mu, \sigma^2)$ ,  $\sigma^2 > 0$ , then the random variable  $V = \frac{(Y-\mu)^2}{\sigma^2} = Z^2$  is a chi-square random variable with 1 degree of freedom. That is,  $Z^2 \sim \chi^2(1)$

## 4.7 Beta Distribution

The Beta distribution is an extremely flexible distribution defined over the closed interval  $0 \leq y \leq 1$ . It is often used to model proportions, such as the proportion of impurities in a chemical product or the proportion of time a machine is under repair. It is very useful in Bayesian inference because it is a “conjugate prior” for many commonly used data distributions (bernoulli, binomial, negative binomial, geometric).

The **Beta distribution** has two parameters,  $\alpha > 0$  and  $\beta > 0$ .

$$f(y) = \frac{y^{\alpha-1}(1-y)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq y < 1,$$

where

$$B(\alpha, \beta) = \int_0^1 y^{\alpha-1}(1-y)^{\beta-1} dy = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

$$E(Y) = \frac{\alpha}{\alpha + \beta}$$

$$V(Y) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

**Example:** A gasoline wholesale distributor has bulk storage tanks that hold fixed supplies and are filled every Monday. Of interest to the wholesaler is the proportion of this supply that is sold during the week. Over many weeks of observation, the distributor found that this proportion could be modeled by a beta distribution with  $\alpha = 4$  and  $\beta = 2$ . Find the probability that the wholesaler will sell at least 90% of her stock in a given week.

## Chapter 4 Group Work

- Let  $Y$  be a continuous random variable with pdf  $f(y) = c(y - y^2)$ ,  $0 < y < 1$ 
  - Find  $c$
  - Find  $F(y)$ .
  - Find  $P(0.3 < Y < 0.6)$
  - Find the median of  $Y$
- The pdf of  $Y$  is given by  $f(y) = \frac{2}{y^3}$ ,  $1 < y < \infty$ . Find  $E(Y)$  and  $V(Y)$ .
- For the function  $f(y) = 4x^c$ ,  $0 < y < 1$ ,
  - Find the constant  $c$  such that  $f(y)$  is a valid pdf
  - Find the cdf
  - Find  $\mu$  and  $\sigma$
  - Find  $P(0.25 < Y < 0.75)$  using the cdf
- Find the mean, variance, and mgf for a continuous uniform distribution.
- The failure of a circuit board interrupts work that utilizes a computing system until a new board is delivered. The delivery time,  $Y$ , is uniformly distributed on the interval one to five days. The cost of a board failure and interruption includes a fixed cost  $c_0$  of a new board and a cost that increases proportionally to  $Y^2$ . If  $C$  is the cost incurred,  $C = c_0 + c_1Y^2$ .
  - Find the probability that the delivery time exceeds two days/
  - In terms of  $c_0$  and  $c_1$ , find the expected cost associated with a single failed circuit board.
- The magnitude of earthquakes recorded in a region of North America can be modeled as having an exponential distribution with mean 2.4, as measured on the Richter scale. Find the probability that an earthquake striking this region will
  - exceed 3.0 on the Richter scale
  - fall between 2.0 and 3.0 on the Richter scale
- Cars arrive at a tollbooth at a mean rate of five cars every ten minutes according to a Poisson process. Find the probability that the toll collector will have to wait longer than 26.30 minutes before collecting the 8th toll.
- The weekly repair cost  $Y$  for a machine has a probability density function given by  $f(y) = 3(1-y)^2$ ,  $0 < y < 1$  with measurements in hundreds of dollars. How much money should be budgeted each week for repair costs so that the actual cost will exceed the budgeted amount only 10% of the time?